FOUR EQUIVALENT CHARACTERISATIONS OF AN ETALE MORPHISM, BY LOCAL AND AFFINE-LOCAL CONDITIONS

THE STACKS PROJECT AUTHORS

Editorial note. All delivered mathematical text is the author's own, copied from the pinned Stacks Project chapter file morphisms.tex; no statement or proof step was rewritten, reordered or removed.

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Editorial note. Both counted spans start at the \begin line of their environment, so the statement and the author's proof are delivered whole; surrounding prose of the section is deliberately outside the counted spans and is carried as standard prerequisites instead.

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Editorial note. The span names the exact reduction (locality of the ring-map property) and the exact definition whose three conditions must be checked, and says where they are checked – 'These all follow from Algebra, Lemma algebra-lemma-etale.' This is a real if short proof of a locality reduction, not a restatement: the fact that the property is local is what has to be established, and the two cards registered next to it (S1-morph-base-change-etale, S1-morph-composition-etale) both rely on it.

Allowed standard prerequisites (editorial).

- The general lemma producing the four equivalent formulations from a local property of ring maps. Lemma \ref{lemma-locally-P}, cited at proof lines 8112-8115. Supplies the whole equivalence as soon as locality is checked.
- The definition of a local property of ring maps, conditions (a), (b) and (c). Definition \ref{definition-property-local}, cited at proof lines 8115-8117. Names the three conditions that have to be verified.
- The algebra lemma on etale ring maps, which verifies those conditions. Algebra, Lemma \ref{algebra-lemma-etale}, cited at proof lines 8117-8118. Discharges all three conditions at once.

Source reference. Original lemma 02GJ: focused-seeds/AGP/sources/stacks-snapshot-20261011/tags.txt, source lines 3230–3230. The lemma's own canonical tag, quoted so that readers can resolve the label inside the delivered text; it points at the counted statement itself and carries no proof content.

Source reference. Original lemma 01ST: focused-seeds/AGP/sources/stacks-snapshot-20261011/tags.txt, source lines 2350–2350. Navigation pointer to a statement the

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counted proof refers to but does not contain; nothing load-bearing is delegated to it.

Source reference. Original lemma 01SR: focused-seeds/AGP/sources/stacks-snapshot-20261011/tags.txt, source lines 2348–2348. Navigation pointer to a statement the counted proof refers to but does not contain; nothing load-bearing is delegated to it.

Source reference. Original lemma 00U2: focused-seeds/AGP/sources/stacks-snapshot-20261011/tags.txt, source lines 1096–1096. Navigation pointer to a statement the counted proof refers to but does not contain; nothing load-bearing is delegated to it.

Lemma 0.2. *Let $f : X \to S$ be a morphism of schemes. The following are equivalent*

1. *The morphism f is étale.*
2. *For every affine opens $U \subset X$, $V \subset S$ with $f(U) \subset V$ the ring map $\mathcal{O}_S(V) \to \mathcal{O}_X(U)$ is étale.*
3. *There exists an open covering $S = \bigcup_{j \in J} V_j$ and open coverings $f^{-1}(V_j) = \bigcup_{i \in I_j} U_i$ such that each of the morphisms $U_i \to V_j$, $j \in J, i \in I_j$ is étale.*
4. *There exists an affine open covering $S = \bigcup_{j \in J} V_j$ and affine open coverings $f^{-1}(V_j) = \bigcup_{i \in I_j} U_i$ such that the ring map $\mathcal{O}_S(V_j) \to \mathcal{O}_X(U_i)$ is étale, for all $j \in J, i \in I_j$.*

Moreover, if f is étale then for any open subschemes $U \subset X$, $V \subset S$ with $f(U) \subset V$ the restriction $f|_U : U \to V$ is étale.

Proof. This follows from Lemma 01ST if we show that the property "$R \to A$ is étale" is local. We check conditions (a), (b) and (c) of Definition 01SR. These all follow from Algebra, Lemma 00U2. □